

Nuclear Astrophysics (NS 3035)

Lec 5: Charged Particle Induced Reactions

The large blue spherical regions represent two positively charged nuclei separated by a Coulomb barrier.

Classically, a charged particle approaching another positively charged nucleus would require enough kinetic energy to overcome the electrostatic repulsion between them. At the temperatures found in stellar interiors, however, most particles do not possess sufficient energy to cross this barrier. The dark region between the nuclei symbolizes the potential barrier created by the Coulomb force. According to classical physics, particles with energies lower than the barrier height should be completely reflected. The colored beam passing through the barrier represents the quantum mechanical wave associated with the incoming charged particle. Unlike a classical particle, a quantum wavefunction does not abruptly vanish at the barrier. Instead, it penetrates into the classically forbidden region and retains a finite amplitude on the opposite side. Consequently, there exists a non-zero probability that the particle will appear beyond the barrier even when its energy is insufficient to overcome it classically.

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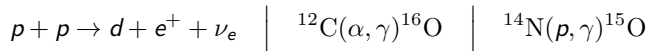
Charged Particle Induced Reactions in Stars

Charged particle induced reactions constitute one of the most important classes of nuclear reactions in stellar environments. These reactions are responsible for both the production of stellar energy and the synthesis of new nuclei. Typical projectiles involved in these reactions are protons (p), alpha particles (α), and occasionally heavier charged ions.

The charged particle reactions

- ▶ Power stars during hydrogen burning, helium burning, and later evolutionary stages
- ▶ Determine the rates of nucleosynthesis responsible for the origin of elements
- ▶ Their reaction rates strongly depend on temperature
- ▶ Their cross sections at stellar energies are extremely small and difficult to measure directly in laboratories
- ▶ Quantum tunneling plays a crucial role in allowing these reactions to occur inside stars

Examples of Charged Particle Reactions



Why Are Reaction Rates Important?

A star continuously evolves throughout its lifetime. As nuclear fuel is consumed, gravitational contraction and thermonuclear burning alter the internal temperature and density (**Recall first 3 classes**). For every stage of stellar evolution we must calculate reaction rates for thousands of nuclear reactions.

Thermally Averaged Reaction Rate

$$\langle \sigma v \rangle = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^{\infty} \sigma(E) E e^{-E/kT} dE$$

[See the notes for additional details and derivations](#)

Meaning of terms

- ▶ $\sigma(E)$: reaction cross section
- ▶ v : relative velocity
- ▶ μ : reduced mass
- ▶ T : stellar temperature

This quantity determines

- ▶ Energy generation rates
- ▶ Evolutionary timescales
- ▶ Element production rates
- ▶ Final chemical composition

The Challenge In Measuring $\sigma(E)$ for Stars

Reaction Rates Depend on Cross Sections

Accurate calculation of stellar reaction rates requires full knowledge of the energy dependent nuclear cross section $\sigma(E)$

Why Is This a Difficult Problem?

Nuclear reactions inside stars occur at very low energies ($\sim$ from a few keV to a few hundred keV)

- ▶ The Coulomb barrier strongly suppresses reaction probabilities
- ▶ Cross sections become extremely small
- ▶ Direct measurements in the laboratory are challenging
- ▶ Experimental signals are often buried beneath background radiation

The Challenge In Measuring $\sigma(E)$ for Stars

Reaction Rates Depend on Cross Sections

Accurate calculation of stellar reaction rates requires full knowledge of the energy dependent nuclear cross section $\sigma(E)$

How Do We Address This Challenge?

- ▶ Cross sections are measured at higher experimentally accessible energies
- ▶ Nuclear theory is used to model the energy dependence of $\sigma(E)$
- ▶ The results are *extrapolated* to the lower energies relevant for stellar interiors

The Central Goal

A major objective of nuclear astrophysics is to determine reliable low-energy cross sections and reaction rates for the conditions found inside stars.

Non-Resonant and Resonant Reactions

The behavior of the reaction cross section determines the classification of nuclear reactions

Non-Resonant Reactions

- ▶ Cross section varies smoothly with energy
- ▶ No sharp structures appear
- ▶ Easier to extrapolate to low energies
- ▶ Typical for many stellar fusion reactions

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Resonant Reactions

- ▶ Cross section exhibits sharp peaks
- ▶ Strong enhancement occurs at specific energies
- ▶ Caused by formation of intermediate compound nuclear states
- ▶ Dominates reaction rates when resonance energies coincide with stellar energies

More Questions

Several key questions arise when studying charged particle induced reactions

- ▶ Can the strong energy dependence of cross sections be removed?
- ▶ At what energies do most stellar reactions occur?
- ▶ Is the reaction probability the same across all energies?
- ▶ What energy window contributes most to stellar energy generation?
- ▶ How can we identify the most effective reaction energy?
- ▶ How does the reaction rate depend on temperature?

Answering these questions leads directly to the concepts of

- ▶ Coulomb barrier
- ▶ Quantum tunneling
- ▶ Astrophysical S-factor
- ▶ Gamow peak
- ▶ Temperature dependent reaction rates

The Coulomb Barrier

Before two nuclei can fuse they must approach closely enough for the **strong nuclear force** to become effective. But, positively charged nuclei repel each other through the **Coulomb force**

$$V_C(r) = \frac{Z_1 Z_2 e^2}{r}$$

where Z_1 = **projectile charge**, Z_2 = **target charge** and r = **separation distance**

Characteristics

- ▶ **Long-range** repulsive interaction.
- ▶ Produces an **energy barrier**.
- ▶ Prevents fusion at **low energies**.

Consequences

- ▶ Fusion probability decreases rapidly.
- ▶ Barrier becomes larger for **heavier nuclei**.
- ▶ Higher kinetic energy is required for fusion.

Only when nuclei approach within the **nuclear radius** does the attractive **strong nuclear force** dominate over the **Coulomb repulsion**, allowing fusion to occur.

Effective Nuclear Potential

The interaction between two nuclei is governed by two competing forces.

Repulsive Interaction

Coulomb Force

- ▶ Acts over long distances.
- ▶ Repels positively charged nuclei.
- ▶ Produces the Coulomb barrier.

Attractive Interaction

Strong Nuclear Force

- ▶ Extremely strong.
- ▶ Short-range interaction.
- ▶ Binds nucleons together.

Effective Nuclear Potential

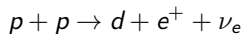
For Fusion to Occur

- ▶ Classically: Projectile energy must exceed the barrier height.
- ▶ Quantum mechanically: nuclei can penetrate the barrier even when $E < V_C$.

This phenomenon is called Quantum Tunneling

The Proton-Proton Fusion Problem

The most important reaction in stellar astrophysics is



For two protons $V_C \approx 550$ keV. The corresponding temperature required for classical barrier crossing is $T \approx 6.4 \times 10^9$ K or $T_9 \approx 6.4$. This leads to a paradox...

- ▶ If stars required such temperatures, proton fusion would occur explosively
- ▶ Stars would burn their fuel almost instantaneously
- ▶ Stable stars like the Sun could not exist

Yet the Sun burns steadily for billions of years. So clearly another mechanism must allow fusion at much lower temperatures

Why Classical Physics Fails

Consider a typical stellar core temperature

$$T = 10^7 \text{ K}$$

or

$$T_9 = 0.01$$

The thermal energy is

$$kT \approx 0.86 \text{ keV}$$

Compare this with the proton-proton Coulomb barrier: $V_C \approx 550 \text{ keV}$

Thus: $kT \ll V_C$ The average proton possesses nearly 600 times less energy than required for classical fusion. Classical mechanics predicts essentially zero fusion probability. Therefore classical physics alone cannot explain stellar energy production.

Maxwell-Boltzmann Distribution is Not Enough

One possible argument is that a few particles in the high-energy tail of the Maxwell-Boltzmann distribution might possess sufficient energy to overcome the barrier

However, the probability ratio is approximately

$$\frac{P(550 \text{ keV})}{P(0.86 \text{ keV})} \approx 10^{-275}$$

This means

- ▶ Particles capable of classical fusion are essentially absent
- ▶ Thermal fluctuations alone cannot explain stellar luminosities
- ▶ Another mechanism must be operating

The observed energy output of stars therefore provides direct evidence for a non-classical process occurring in stellar interiors

Gamow's Solution → Quantum Tunneling

The Classical Problem

Charged particles inside stars possess energies far below the Coulomb barrier. Classically, fusion should therefore be impossible.

Quantum Mechanical Solution

George Gamow proposed that

- ▶ Particles behave as matter waves
- ▶ The wavefunction extends into the forbidden region
- ▶ A particle can penetrate the Coulomb barrier even when $E < V_C$
- ▶ This process is called **quantum tunneling**
- ▶ Once inside the nuclear radius, the strong force can cause fusion

Quantum tunneling allows nuclear reactions to occur at stellar temperatures far below those required for classical barrier penetration

Gamow's Solution → Quantum Tunneling

Physical Picture

- ▶ A particle does **not** need energy greater than the barrier height
- ▶ The quantum wavefunction extends into the classically forbidden region
- ▶ A finite probability exists for the particle to appear on the other side of the barrier.
- ▶ Once inside the nuclear radius, the strong force can trigger fusion

$$P_{\text{tunnel}} \propto e^{-2G}$$

where G is the **Gamow factor**

Typical Tunneling Probabilities

$$E = 1 \text{ keV} \Rightarrow P \sim 10^{-10}$$

$$E = 2 \text{ keV} \Rightarrow P \sim 10^{-7}$$

$$E = 5 \text{ keV} \Rightarrow P \sim 10^{-4}$$

Remark

Even these tiny probabilities are enormously larger than the classical prediction

$$P_{\text{classical}} \approx 10^{-275}$$

Importance of Quantum Tunneling in Stars

Without Tunneling

- ▶ Fusion requires $T \sim 10^9$ K
- ▶ Extremely high reaction rates
- ▶ Stars would burn fuel rapidly
- ▶ Long stellar lifetimes impossible
- ▶ Stable stars could not exist

With Tunneling

- ▶ Fusion occurs near $T \sim 10^7$ K
- ▶ Controlled energy generation
- ▶ Gradual fuel consumption
- ▶ Lifetimes of billions of years
- ▶ Continuous nucleosynthesis

Take Home Message

The existence of stars, planets, heavy elements, and ultimately life depends on the quantum mechanical tunneling of charged particles through the Coulomb barrier.

Summary

- ▶ Charged particle induced reactions are the primary energy source in stars
- ▶ Reaction rates depend strongly on the nuclear cross section and stellar temperature
- ▶ Non-resonant reactions exhibit smooth energy dependence of the cross section
- ▶ Positively charged nuclei face a Coulomb barrier that suppresses fusion
- ▶ Proton-proton fusion requires approximately 550 keV classically
- ▶ Typical stellar temperatures provide only ~ 1 keV thermal energies
- ▶ Maxwell-Boltzmann statistics alone cannot explain stellar fusion
- ▶ Quantum tunneling provides a finite barrier penetration probability
- ▶ Gamow's theory explains how fusion occurs at stellar temperatures

The existence of stars is one of the most spectacular manifestations of quantum mechanics on a cosmic scale.

Outline

Charged Particle Induced Reactions

Gamow Peak Width, Temperature Dependence of Reaction Rates, and Electron Screening

The Need for an Astrophysical S-Factor

Charged-particle reaction cross sections depend very strongly on energy, mainly because of the Coulomb repulsion between the interacting nuclei. As the energy drops below the Coulomb barrier, the barrier-penetration probability falls off exponentially, so the cross section becomes extremely small.

In stars, most reactions occur at only a few keV to a few hundred keV – far below the Coulomb barrier. As a result:

- ▶ Cross sections are exceedingly small
- ▶ Direct laboratory measurement at stellar energies is essentially impossible
- ▶ Experimental uncertainties grow rapidly as energy decreases
- ▶ Stellar rates must be extrapolated from higher-energy laboratory data

The central challenge is therefore to separate the rapidly varying Coulomb-barrier effects from the slowly varying nuclear-structure effects. This motivates the **Astrophysical S-factor** – one of the most important quantities in stellar reaction theory.

Coulomb Tunneling: Setting Up the Problem

For charged-particle reactions in stars, the projectile energy is far below the Coulomb barrier, $E \ll V_C$. Classically fusion cannot occur; quantum mechanically the particles can tunnel through the classically forbidden region.

To estimate the tunneling probability we assume:

- ▶ relative motion is a two-body problem with reduced mass μ ;
- ▶ outside the nuclear radius the interaction is purely Coulombic, $V(r) = \frac{Z_1 Z_2 e^2}{r}$;
- ▶ in the forbidden region $E < V(r)$;
- ▶ the WKB approximation is valid (wavefunction varies slowly compared to its wavelength);
- ▶ nuclear forces act only once the Coulomb barrier has been penetrated.

These assumptions lead to a tunneling probability $P(E) \propto e^{-2\pi\eta}$, where $\eta = Z_1 Z_2 e^2 / \hbar v$ is the **Sommerfeld parameter** – derived on the next slide.

WKB Derivation of the Gamow Factor

Starting from the radial Schrödinger equation,

$$-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dr^2} + V(r)u = Eu,$$

the classically forbidden region satisfies $V(r) > E$. In the WKB approximation the barrier transmission probability is

$$P(E) \simeq \exp \left[-2 \int_{r_N}^{r_c} \kappa(r) dr \right], \quad \kappa(r) = \frac{\sqrt{2\mu [V(r) - E]}}{\hbar}.$$

WKB Derivation of the Gamow Factor

For the Coulomb potential $V(r) = Z_1 Z_2 e^2 / r$, the outer turning point satisfies $V(r_c) = E$, so

$$r_c = \frac{Z_1 Z_2 e^2}{E}.$$

Since the nuclear radius satisfies $r_N \ll r_c$, the WKB integral reduces to

$$\int_{r_N}^{r_c} \kappa(r) dr \approx \pi \eta \quad \Longrightarrow \quad P(E) = e^{-2\pi \eta}.$$

WKB Derivation of the Gamow Factor

Defining the Sommerfeld parameter

$$\eta = \frac{Z_1 Z_2 e^2}{\hbar v},$$

gives the famous **Gamow factor**

$$G(E) = e^{-2\pi\eta}.$$

Using $v = \sqrt{2E/\mu}$,

$$2\pi\eta = 31.29 Z_1 Z_2 \sqrt{\frac{\mu}{E}} \quad (E \text{ in keV, } \mu \text{ in amu}).$$

The Gamow factor is the exponential suppression of low-energy charged-particle fusion due to Coulomb-barrier penetration.

Cross Section for Non-Resonant Reactions

For a non-resonant charged-particle reaction, the cross section is written

$$\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta}.$$

This separates the physics into three pieces:

1. $1/E$: the geometric, de Broglie-wavelength dependence of the reaction probability;
2. $e^{-2\pi\eta}$: Coulomb-barrier penetration by quantum tunneling;
3. $S(E)$: the intrinsic nuclear structure information and matrix elements.

Unlike the steep exponential Gamow factor, $S(E)$ varies slowly with energy for non-resonant reactions – which makes it far easier to extrapolate measured data toward astrophysical energies.

Definition and Physical Meaning of the S-Factor

The astrophysical S-factor is defined by inverting the cross-section formula:

$$S(E) = \sigma(E) E e^{2\pi\eta}.$$

It removes the two dominant sources of energy dependence – the geometric $1/E$ term and the Coulomb tunneling term $e^{-2\pi\eta}$ – leaving a quantity that reflects mainly:

- ▶ nuclear structure effects;
- ▶ spectroscopic properties;
- ▶ overlap of initial- and final-state nuclear wavefunctions;
- ▶ transition matrix elements.

For non-resonant reactions $S(E)$ is slowly varying and is often approximated by a low-order polynomial,

$$S(E) = S(0) + S'(0) E + \frac{1}{2} S''(0) E^2 + \dots,$$

making it the key tool for stellar reaction-rate calculations.

Why the S-Factor Improves Extrapolation

Experimental measurements are rarely performed exactly at stellar energies:

- ▶ stellar fusion typically occurs between 10 and 300 keV;
- ▶ laboratory measurements are usually made at significantly higher energies;
- ▶ direct low-energy measurements are difficult because cross sections fall exponentially.

Extrapolating $\sigma(E)$ directly is unreliable: small energy uncertainties translate into enormous uncertainties because of its exponential dependence.

After removing the Coulomb suppression, $S(E) = \sigma(E) E e^{2\pi\eta}$ varies far more smoothly, so:

- ▶ extrapolation becomes more reliable and uncertainties shrink;
- ▶ reaction rates are computed more accurately;
- ▶ stellar evolution models become more trustworthy.

Reaction Rate in Terms of the S-Factor

The thermally averaged reaction rate per particle pair is

$$\langle \sigma v \rangle = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty E \sigma(E) e^{-E/kT} dE.$$

Substituting $\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta}$ gives

$$\langle \sigma v \rangle = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty S(E) e^{-2\pi\eta} e^{-E/kT} dE.$$

The rate is governed by the product of two competing exponentials, $e^{-2\pi\eta}$ and $e^{-E/kT}$. Their competition produces one of the central concepts of nuclear astrophysics: the **Gamow Peak**.

Origin of the Gamow Peak

Two opposing effects determine the probability that a fusion reaction occurs.

Maxwell–Boltzmann factor $e^{-E/kT}$: favours *low* energies, since high-energy particles are rare.

Tunneling probability $e^{-2\pi\eta}$: favours *high* energies, since barrier penetration becomes easier.

- ▶ Low energies: many particles, but poor tunneling.
- ▶ High energies: efficient tunneling, but very few particles.

The product $e^{-E/kT} e^{-2\pi\eta}$ has a maximum at an intermediate energy. This maximum defines the **Gamow Peak** – the energy window where most stellar nuclear reactions actually occur.

Physical Interpretation of the Gamow Peak

The Gamow Peak answers a key question:

At what energy do most stellar fusion reactions occur?

The answer is *not* $E = kT$. Although the Maxwell–Boltzmann distribution peaks at kT , fusion occurs preferentially at higher energies because tunneling becomes more effective there.

The centroid of the Gamow Peak, E_0 , satisfies $E_0 > kT$.

Stellar reactions therefore occur mainly in the high-energy tail of the Maxwell–Boltzmann distribution, not at the thermal peak itself – one of the central insights of nuclear astrophysics.

Gamow Peak Energy

Maximizing the integrand of the reaction-rate expression gives the most effective energy for charged-particle fusion:

$$E_0 = 1.22 (Z_1^2 Z_2^2 \mu T_6^2)^{1/3} \text{ keV},$$

where Z_1, Z_2 are the nuclear charges, μ is the reduced mass in amu, and T_6 is the temperature in units of 10^6 K.

Key conclusions:

- ▶ E_0 increases with temperature and with nuclear charge;
- ▶ heavier, more highly charged nuclei require higher effective reaction energies;
- ▶ even so, E_0 remains well below the full Coulomb barrier for every case of interest.

Quantum tunneling therefore remains essential for stellar fusion at all stages.

Examples of Gamow Peak Energies

$$p + p : \quad E_0 \approx 5.9 \text{ keV}$$

$$p + {}^{14}\text{N} : \quad E_0 \approx 26.5 \text{ keV}$$

$$\alpha + \alpha : \quad E_0 \approx 56 \text{ keV}$$

$${}^{12}\text{C} + \alpha : \quad E_0 \approx 237 \text{ keV}$$

These examples show the general trend $E_0 \uparrow$ as $Z_1 Z_2 \uparrow$: larger nuclear charges create stronger Coulomb barriers.

Nevertheless $E_0 < V_C$ for essentially every stellar fusion reaction – confirming that fusion in stars is fundamentally a quantum-mechanical phenomenon.

Outline

Charged Particle Induced Reactions

Gamow Peak Width, Temperature Dependence of Reaction Rates, and Electron Screening

Recap: The Gamow Peak

The reaction rate for non-resonant charged-particle reactions is governed by the product of two competing factors: $e^{-E/kT}$ (Maxwell–Boltzmann, favours low energy) and $e^{-2\pi\eta}$ (Coulomb tunneling, favours high energy).

Their product produces a narrow peak, the **Gamow Peak**, centred at the most effective reaction energy:

$$E_0 = 1.22 (Z_1^2 Z_2^2 \mu T_6^2)^{1/3} \text{ keV},$$

where Z_1, Z_2 are the projectile and target charges, μ is the reduced mass in amu, and T_6 is the temperature in units of 10^6 K.

The Gamow peak identifies the narrow energy range responsible for most stellar fusion reactions – our goal now is to quantify its height, width, and the consequences for reaction rates.

Effect of Nuclear Charge on the Gamow Peak

The location of the Gamow peak depends strongly on the Coulomb barrier. As $Z_1 Z_2$ increases, the Coulomb repulsion grows stronger, so:

- ▶ the centroid energy E_0 shifts to higher energies;
- ▶ the tunneling probability decreases;
- ▶ the area under the Gamow peak shrinks;
- ▶ the overall reaction probability falls.

$$E_0(p + p) < E_0(p + {}^{12}\text{C}) < E_0({}^{12}\text{C} + \alpha)$$

Reactions with the smallest Coulomb barriers therefore dominate the *earliest* stages of stellar evolution – this single trend is the seed of the whole burning sequence we build up over the next few slides.

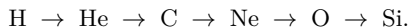
Physical Meaning of the Gamow Peak Area

The area under the Gamow peak is the total probability that a reaction occurs in the stellar plasma – it shrinks rapidly as V_C increases.

Astrophysical consequence:

- ▶ nuclei with the smallest Coulomb barriers are consumed first;
- ▶ more highly charged nuclei stay largely unburned until much higher temperatures are reached.

This is why stellar burning proceeds in a strict sequence of progressively higher ignition temperatures:



Gaussian Approximation to the Gamow Peak

The exact Gamow peak is not perfectly Gaussian, but near its maximum it is very well approximated by one. The integrand $e^{-2\pi\eta}e^{-E/kT}$ can be written near E_0 as

$$I(E) = I_{\max} \exp \left[- \left(\frac{E - E_0}{\Delta/2} \right)^2 \right].$$

This approximation is useful because it allows:

- ▶ analytical integration of the reaction-rate integral;
- ▶ a closed-form temperature dependence for the reaction rate;
- ▶ a direct expression for the width of the Gamow peak;
- ▶ a simpler physical picture of stellar burning conditions.

Maximum Height of the Gamow Peak

The integrand reaches its maximum at $E = E_0$, with peak height

$$I_{\max} = \exp\left(-\frac{3E_0}{kT}\right).$$

The peak height therefore depends *exponentially* on the Coulomb barrier through E_0 : as E_0 increases, $I_{\max} \propto e^{-3E_0/kT}$ falls dramatically.

Even a modest increase in Coulomb barrier can reduce the reaction probability by many orders of magnitude – this is why stellar fusion rates are extraordinarily sensitive to both temperature and nuclear charge.

Width of the Gamow Peak

Matching the curvature of the exact integrand to the Gaussian approximation gives the width

$$\Delta = \frac{4}{\sqrt{3}} \sqrt{E_0 kT} \quad \text{or numerically} \quad \Delta = 0.749 (Z_1^2 Z_2^2 \mu T_6^5)^{1/6} \text{ keV}.$$

Key points:

- ▶ $\Delta < E_0$ always;
- ▶ the width grows with both temperature and Coulomb barrier;
- ▶ heavier, more highly charged nuclei react over a broader energy window.

The width sets the energy range over which most stellar reactions actually occur.

The Gamow Window

Most stellar fusion reactions occur within the interval

$$E_0 - \frac{\Delta}{2} \leq E \leq E_0 + \frac{\Delta}{2},$$

known as the **Gamow Window**. It is central to experimental nuclear astrophysics because it fixes:

- ▶ the energy range relevant inside stars;
- ▶ the target energies for laboratory measurements;
- ▶ the energies at which the reaction rate must be known accurately.

The Gamow window often lies below what laboratories can measure directly – which is exactly why S-factor extrapolation (covered last lecture) is essential.

Putting It Together: the Burning Sequence

The increasing Coulomb barrier of successive fuels, combined with the exponential sensitivity of $I_{\max}$, explains the staged structure of stellar evolution:

- ▶ hydrogen burning ignites first (smallest barrier) and dominates early energy production;
- ▶ once hydrogen is exhausted, the core contracts and heats until helium burning becomes possible, restoring stability;
- ▶ the same contract-heat-ignite cycle repeats for carbon, neon, oxygen, and silicon burning, each requiring a higher core temperature.

In short: stellar evolution is paced by the rising Coulomb barriers of successive nuclear fuels.

Temperature Dependence of the Reaction Rate

Combining the Gaussian approximation, a constant S-factor, and the expressions for E_0 and Δ gives the closed-form reaction rate

$$\langle \sigma v \rangle = 7.2 \times 10^{-19} \frac{S_0}{\mu Z_1 Z_2} \tau^2 e^{-\tau} \text{ cm}^3 \text{ s}^{-1}, \quad \tau = \frac{3E_0}{kT}.$$

This single formula captures the extreme temperature sensitivity of charged-particle fusion: even a small rise in T produces an enormous change in $\langle \sigma v \rangle$.

Reaction Rate Sensitivity to Temperature

Near a given operating temperature, the reaction rate behaves locally as a power law, $\langle\sigma v\rangle \propto T^n$, with n set by $\tau = 3E_0/kT$:

$$n \sim 4 \text{ for } p + p \text{ burning,} \quad n \gg 10 \text{ for heavier charged-particle reactions.}$$

Consequence – a stellar thermostat:

- ▶ a small temperature rise produces a large rise in fusion rate and energy output;
- ▶ the extra energy drives expansion of the stellar material;
- ▶ expansion cools the core, reducing the rate again.

This negative-feedback loop is part of why stars remain stable for billions of years.

Electron Screening Effect

Up to now we assumed reactions between bare nuclei. In a laboratory target, however, nuclei are surrounded by atomic electrons, which partially screen the nuclear charge.

As a result:

- ▶ the projectile experiences a reduced Coulomb barrier;
- ▶ barrier penetration becomes easier;
- ▶ the measured cross section is slightly enhanced relative to the bare-nucleus value.

Effectively, screening adds a small energy shift to the incident particle, increasing its tunneling probability.

Magnitude of the Screening Correction

The characteristic screening energy is

$$U_e = \frac{Z_1 Z_2 e^2}{R_a},$$

with R_a the atomic radius.

For laboratory measurements the correction is usually small, since Gamow-window energies are typically much larger than U_e – data can often be treated as bare-nucleus reactions. For $p + p$:

- ▶ screening enhancement $\approx 33\%$ at 1 keV;
- ▶ $\approx 2.3\%$ at 5 keV;
- ▶ falling rapidly at higher energies.

Screening is therefore usually negligible for most laboratory astrophysics measurements.

Electron Screening in Stellar Matter

Although small in the laboratory, screening can become important inside dense stellar plasmas. At very high densities, many electrons surround each nucleus, effectively reducing the Coulomb barrier and easing penetration, which increases the fusion rate.

The screened rate is written as

$$\langle\sigma v\rangle_{\text{screened}} = f\langle\sigma v\rangle_{\text{bare}}, \quad 1 < f < 2$$

for many astrophysical environments.

Electron screening therefore generally *enhances* stellar reaction rates, and its importance grows with increasing plasma density.

Summary (1/2): The Gamow Peak

- ▶ The Gamow peak defines the energy range where most stellar fusion reactions occur.
- ▶ Increasing nuclear charge shifts the peak to higher energy and shrinks its area.
- ▶ Near its maximum, the peak is well approximated by a Gaussian.
- ▶ Peak height: $I_{\max} = \exp(-3E_0/kT)$.
- ▶ Peak width: $\Delta = \frac{4}{\sqrt{3}}\sqrt{E_0 kT}$.
- ▶ Gamow window: $E_0 - \frac{\Delta}{2} \leq E \leq E_0 + \frac{\Delta}{2}$.

Summary (2/2): Temperature Dependence & Screening

- ▶ Reaction rates are extremely sensitive to temperature, $\langle \sigma v \rangle \propto T^n$ with n often $\gg 1$.
- ▶ This sensitivity, paired with rising Coulomb barriers, sets the sequence of stellar burning stages: $\text{H} \rightarrow \text{He} \rightarrow \text{C} \rightarrow \text{Ne} \rightarrow \text{O} \rightarrow \text{Si}$.
- ▶ Electron screening lowers the effective Coulomb barrier and enhances fusion probability.
- ▶ Screening is usually small in laboratory measurements but can become significant in dense stellar plasmas.